Pediatric Immunology

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

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Objectives

1. Basic concepts of recognizing Pediatric Immune Deficiency Disorders (PIDD)
2. Common immunodeficiencies
   - Phagocyte disorder: Lymphocyte Adhesion Deficiency (LAD)
   - Combined T and B disorder: Severe Combined Immune Deficiency (SCID)
   - T cell disorder: DiGeorge Syndrome
   - B cell disorder: Bruton X linked Agammaglobinemia, Hyper IgM, Job syndrome
   - Other: Wiskott-Aldrich Syndrome (WAS), Ataxia-telangiectasia (AT)
3. Resources
4. Summarization

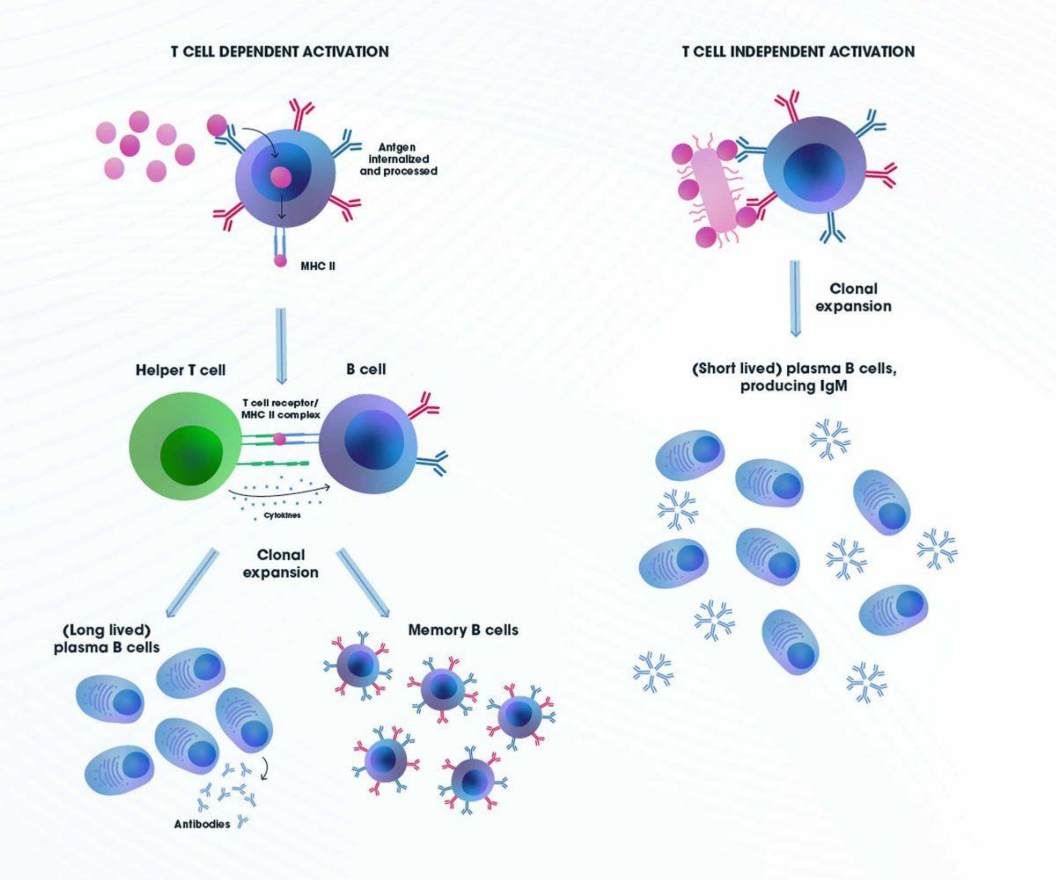
Starter Question

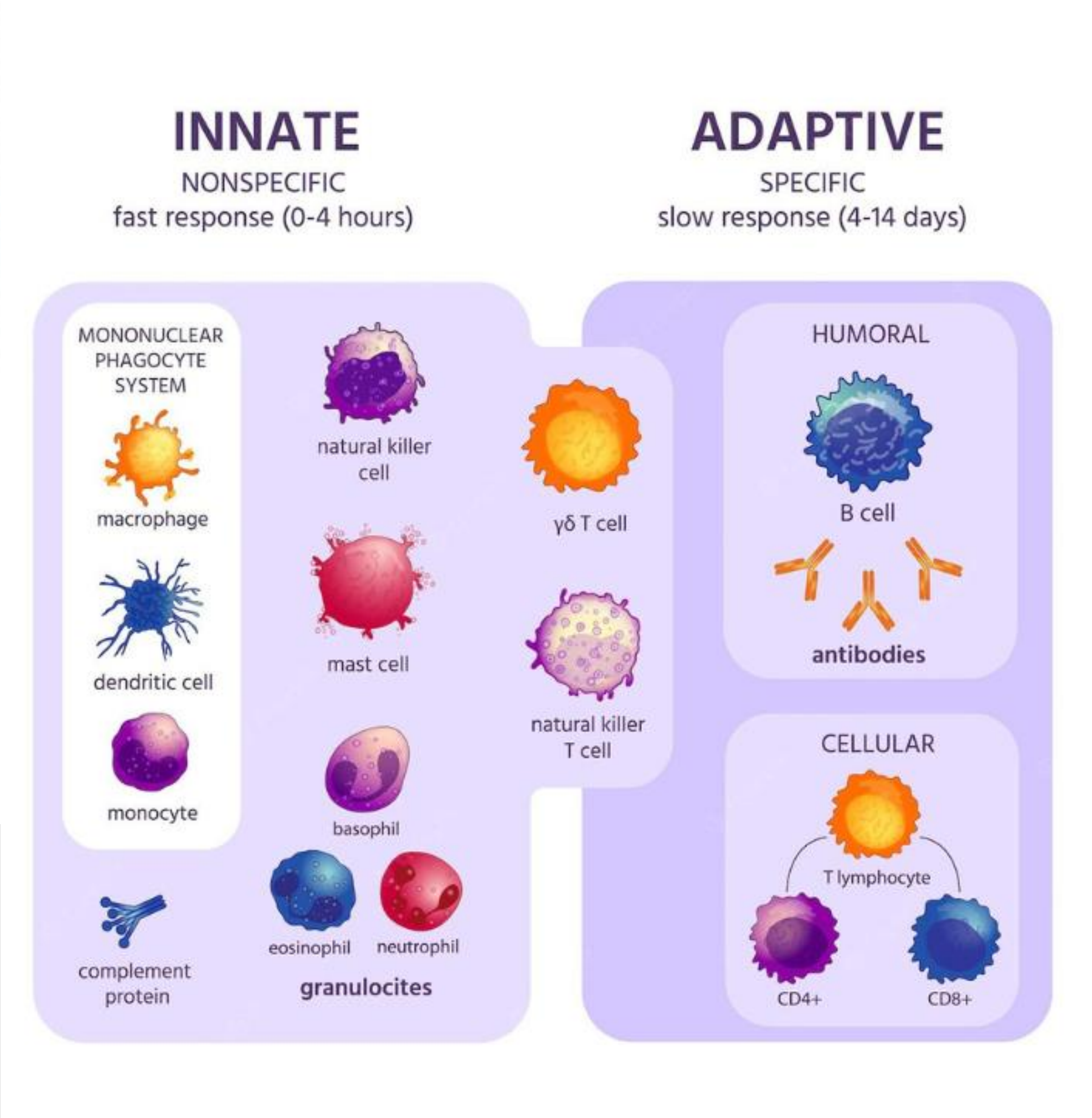
- A 2 month old M infant is brought into your office with a recent hospital admission for 2 weeks for Pneumocystis (PCP) Pneumonia requiring respiratory support, in reviewing the chart, the infant has had multiple episodes of thrush that required prolonged antifungal treatment. Growth chart shows the infants head circumference, weight and length are below the <1.5%tile, on exam, he has severe seborrheic dermatitis and a bulging R TM. You review the xray from the hospital and they comment on an absent thymic shadow. Flow Cytometry shows decreased B, T and NK cells and decreased immunoglobulins. What is the most likely mechanism causing this presentation?

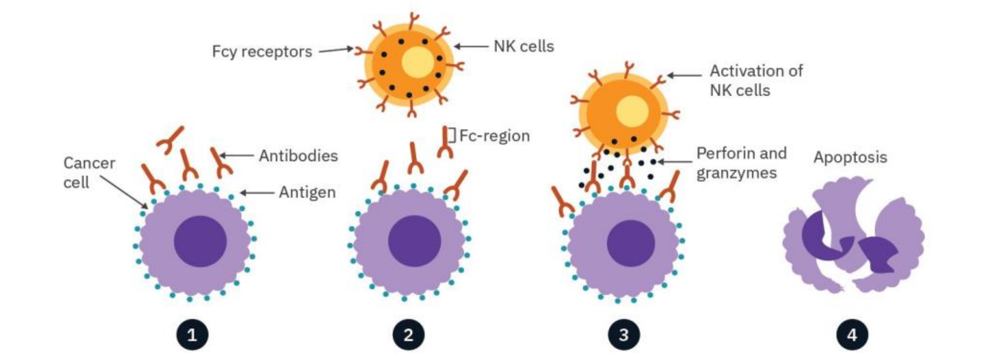
Starter Question

- A 2 month old M infant is brought into your office with a recent hospital admission for 2 weeks for Pneumocystis (PCP) Pneumonia requiring respiratory support, in reviewing the chart, the infant has had multiple episodes of thrush that required prolonged antifungal treatment. Growth chart shows the infants head circumference, weight and length are below the <1.5%tile, on exam, he has severe seborrheic dermatitis and a bulging R TM (tympanic membrane). You review the xray from the hospital and they comment on an absent thymic shadow. Flow Cytometry shows decreased B, T and NK cells and decreased immunoglobulins. What is the most likely mechanism causing this presentation?

Basic concepts of immunodeficiencies







Antibody mediated cytotoxicity

Basic concepts of immunodeficiencies

TABLE 11-7

Types of Infections Associated with Specific Immune Cell Deficiencies

	Cell Type	Types of Infections	Typical Infectious Organisms
Adaptive immune system	B cell	Sinopulmonary infections	*Streptococcus pneumoniae, Haemophilus influenzae, Staphylococcus aureus, Pseudomonas, Enterovirus, Giardia*
	T cell	Opportunistic infections	*Candida, Pneumocystis, Mycobacteria, cytomegalovirus*
Innate immune system	Phagocyte	Abscesses, soft tissue infections	*Aspergillus, Staphylococcus aureus, coagulase-negative staphylococci, Serratia marcescens*
	Complement	Sepsis, meningitis	*Neisseria meningitidis*

TABLE 11-6

Typical Age of Onset of Primary Immunodeficiencies

0–6 months	SCID DiGeorge syndrome Leukocyte adhesion defects
6 months–2 years	Antibody deficiencies (X-linked agammaglobulinemia and hyper-IgM syndrome)
2–6 years	Less severe antibody deficiencies (IgA deficiency and selective IgG antibody deficiency)
6–18 years	Common variable immunodeficiency Secondary immunodeficiencies

Ig, immunoglobulin; SCID, severe combined immunodeficiency.

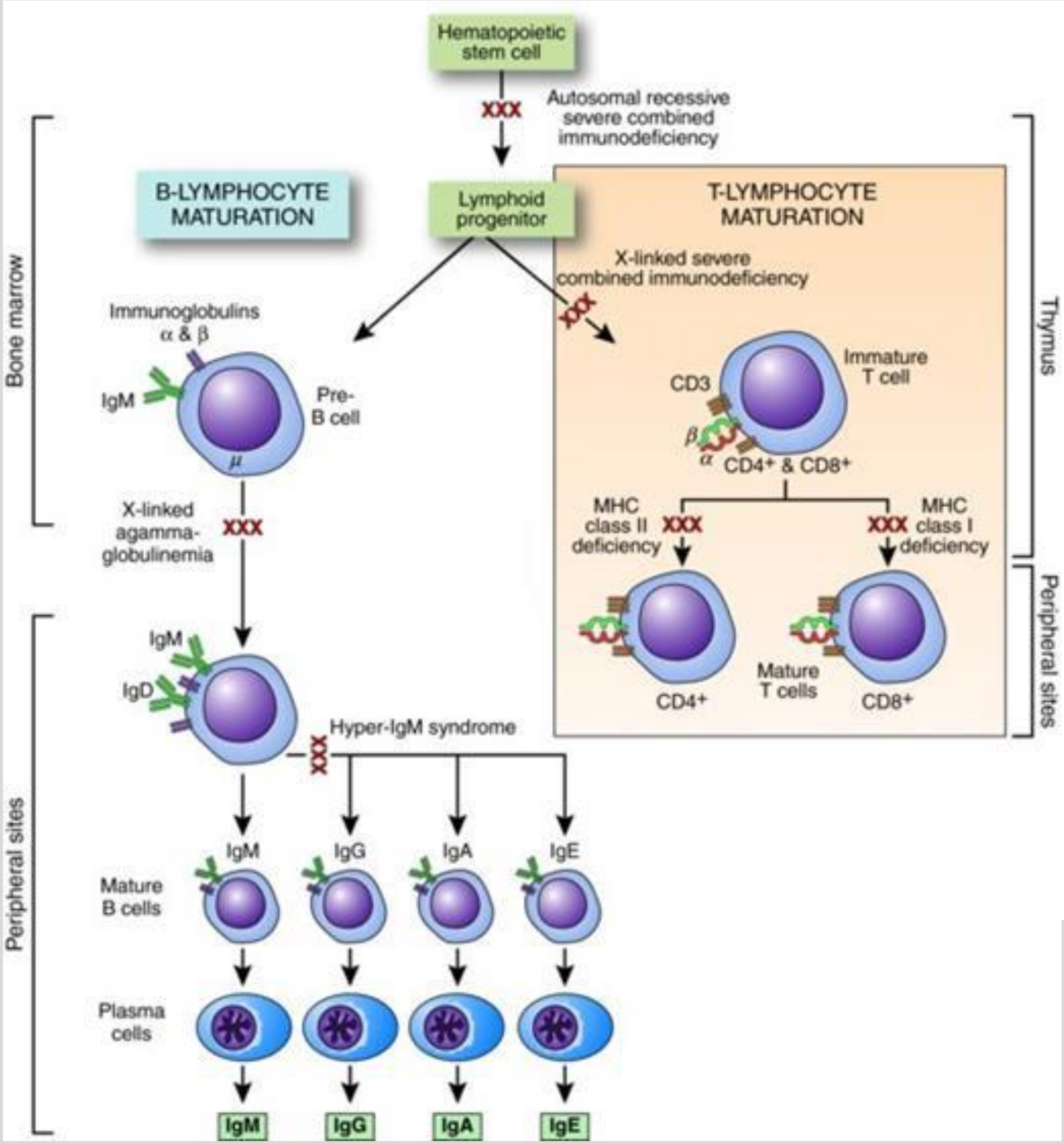
TABLE 11-5

Mendelian Inheritance of Primary Immunodeficiencies

X-linked	Chronic granulomatous disease Wiskott-Aldrich syndrome X-linked agammaglobulinemia X-linked hyper-IgM syndrome X-linked SCID (more common)
Autosomal recessive	Ataxia telangiectasia Leukocyte adhesion defect Chédiak-Higashi syndrome Shwachman-Diamond syndrome Autosomal recessive SCID
Autosomal dominant	Job (hyper-IgE) syndrome Hereditary angioedema
Sporadic	DiGeorge syndrome Common variable immunodeficiency

Ig, immunoglobulin; SCID, severe combined immunodeficiency.

Allergic and Immunologic Disorders | Step-up to Pediatrics, 1e | Clerkship | Health Library



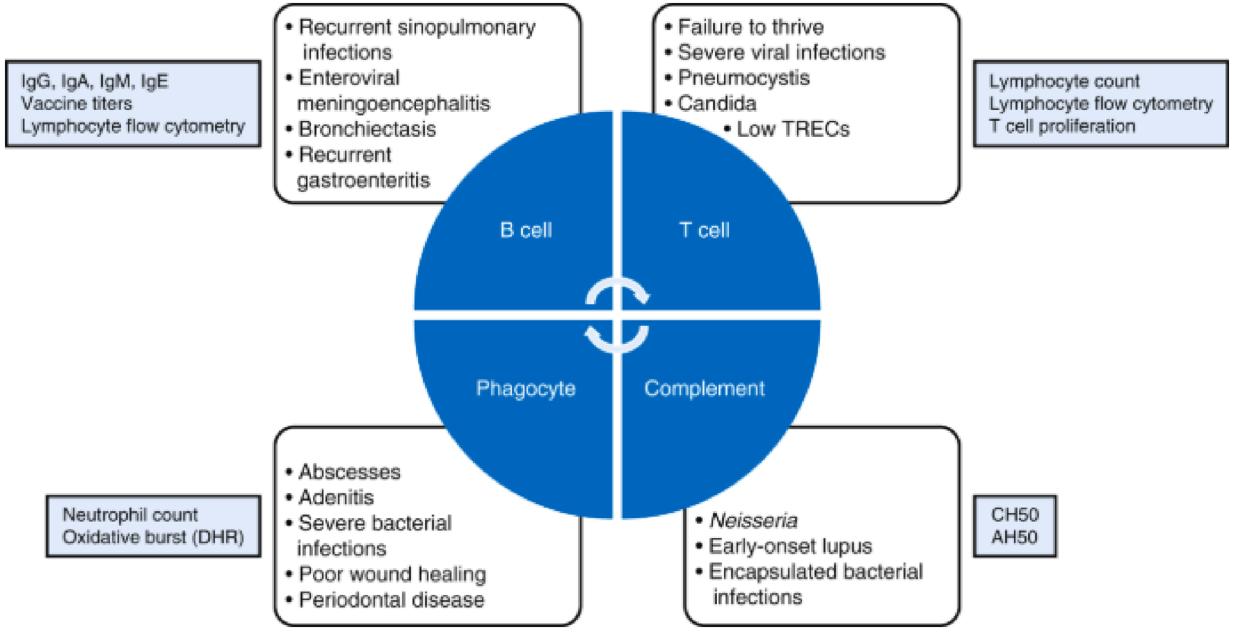


FIG. 164.1

When infections are the major manifestation of the inborn error of immunity, several types of laboratory tests can yield a diagnosis. The specific type of infection suggests the effector arm of the immune system that may be dysfunctional, and the testing can be applied in a targeted fashion. Laboratory evaluation is noted in the light blue boxes. DHR, dihydrorhodamine, Ig, immunoglobulin; TRECs, T cell receptor excision circles.

Nelson's Pediatrics

Basic concepts of immunodeficiencies

[Immunopathology | Rubin's Pathology: Mechanisms of Human Disease, 8e | Premium Basic Sciences | Health Library]

Primary immune deficiency disorders (PIDD)

- 550 and counting different disorders
- Diagnosis tough because
  - Low index of suspicion
  - Low disease incidence
  - Infections, cutaneous lesions, failure to thrive, autoimmunity (can present similar to non specific viral illnesses, or allergies etc) require coordination of care
  - Varied presentations
  - Complex phenotypes
  - Preschool children normally get 6-10 viral infections a year
- Exceed this amt, or severity of symptoms or very prolonged duration – indicating trouble clearing pathogen
- Usually requiring hospitalization for severe symptoms
- Prolonged or ineffective IV abx
- Unusual pathogens (opportunistic infections) or infections from live attenuated vax (rotavirus, VZV, MMR)
- Neutropenia/Lymphocytopenia may be present
- Family hx of PIDD or consanguinity
- Immune dysregulation: autoimmune disease or autoinflammatory conditions (early onset SLE, very early onset IBD, fever syndromes)

Basic concepts of immunodeficiencies

- History and physical exam
- CBC with differential counts
- Cultures/Imaging
- IgG, IgA, IgM, IgE levels
- Diphtheria, tetanus titers
- Pneumococcus, *Haemophilus* titers
- CH50, AH50

If normal, consult with clinical immunologist, consider the following

Suspect phagocyte defect

- Dihydrorhodamine for CGD
- Flow cytometry for LAD1 (CD11/CD18) or LAD2 (CD15)
- Consider neutrophil chemotaxis assays
- IL-12 receptor/IFNγ receptor expression and function

Suspect humoral defect

- Flow cytometry for T cells (CD4, CD8), B cells, NK cells
- Enumerate percentage of naïve, memory, and switched memory B cells
- Functional B cell studies

Suspect combined defect

- Flow cytometry for T cells (CD4, CD8), B cells, NK cells
- Enumerate percentage of memory vs naïve T cells (CD45 isoform expression)
- T cell mitogen studies

Nelson's Pediatrics

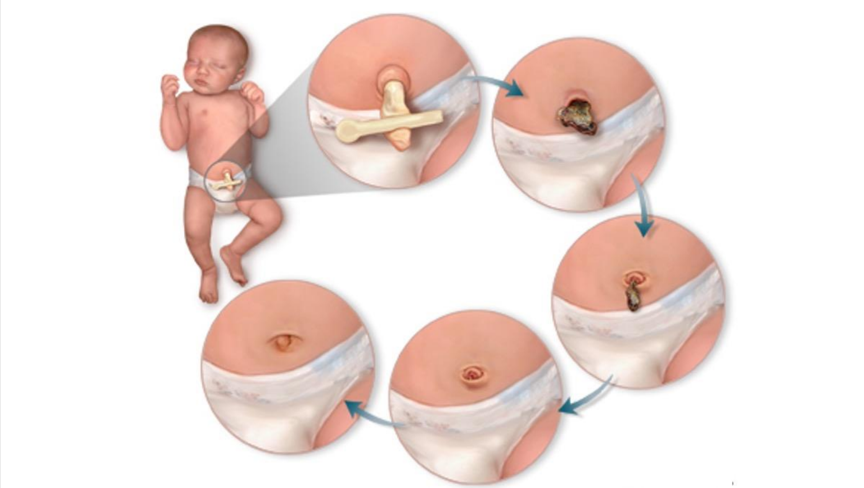
What are we covering?

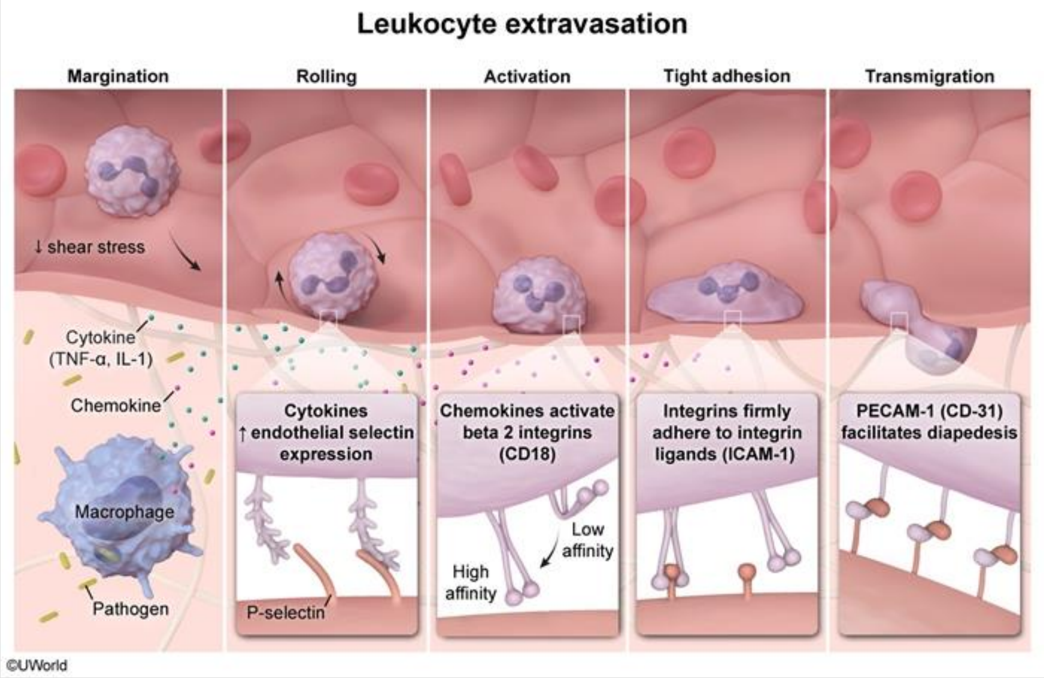
- NOT covered in this lecture
- IgA deficiency
- CVID
- Chronic Mucocutaneous candidiasis
- Chediak-Higashi syndrome
- Chronic Granulomatous disease
- Complement deficiency

- Covered in this lecture
- Phagocyte disorder: LAD
- Combined T and B disorder: SCID
- T cell disorder: DiGeorge Syndrome
- B cell disorder: Bruton X linked Agammaglobinemia, Hyper IgM, Job syndrome
- Other: WAS, Ataxia

Phagocyte disorder: Leukocyte Adhesion Deficiency (type 1)

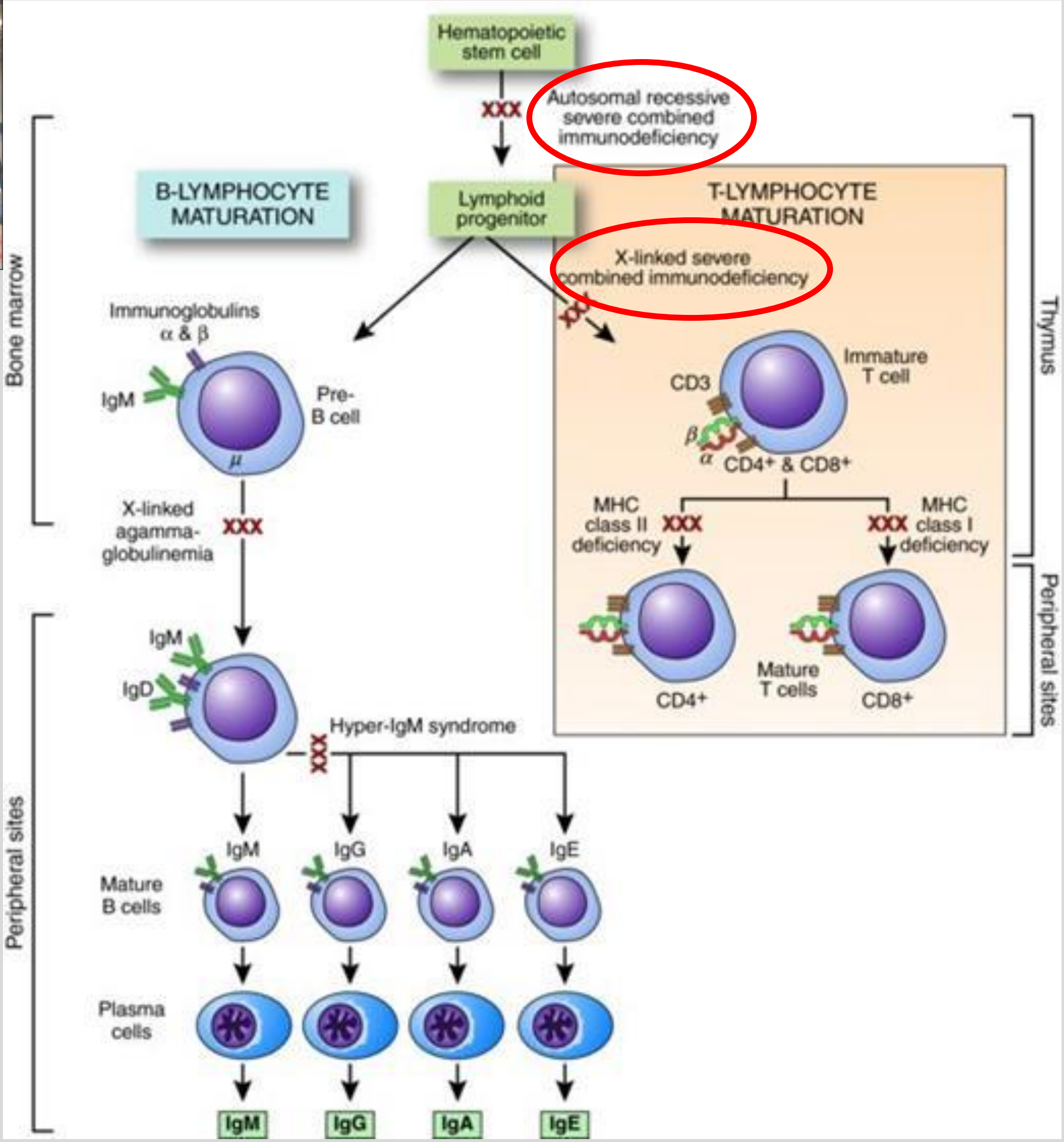
- Autosomal recessive
- Defect in beta-2 integrin (CD18) protein
  - Also called Leukocyte function-associated antigen type 1 (LFA-1)
- Mostly affecting Phagocytes and chemotaxis and migration to site of need
- Clinically:
  - Recurrent skin and mucosal bacterial infections
  - Absent PUS (phagocytic cells can't get to area of need causing inflammation, walling off etc)
  - Impaired wound healing
  - Delayed separation of umbilical cord (normally 14 days, delayed >30 days)
  - Risk for Omphalitis and sepsis
  - Severe gingivitis
- Labs:
  - Neutrophilia >20k
  - Absence of neutrophils at infection site
- Treatment:
  - Ppx Abx
  - Stem cell transplant





Combined T and B cell disorder: Severe combined immunodeficiency (SCID)

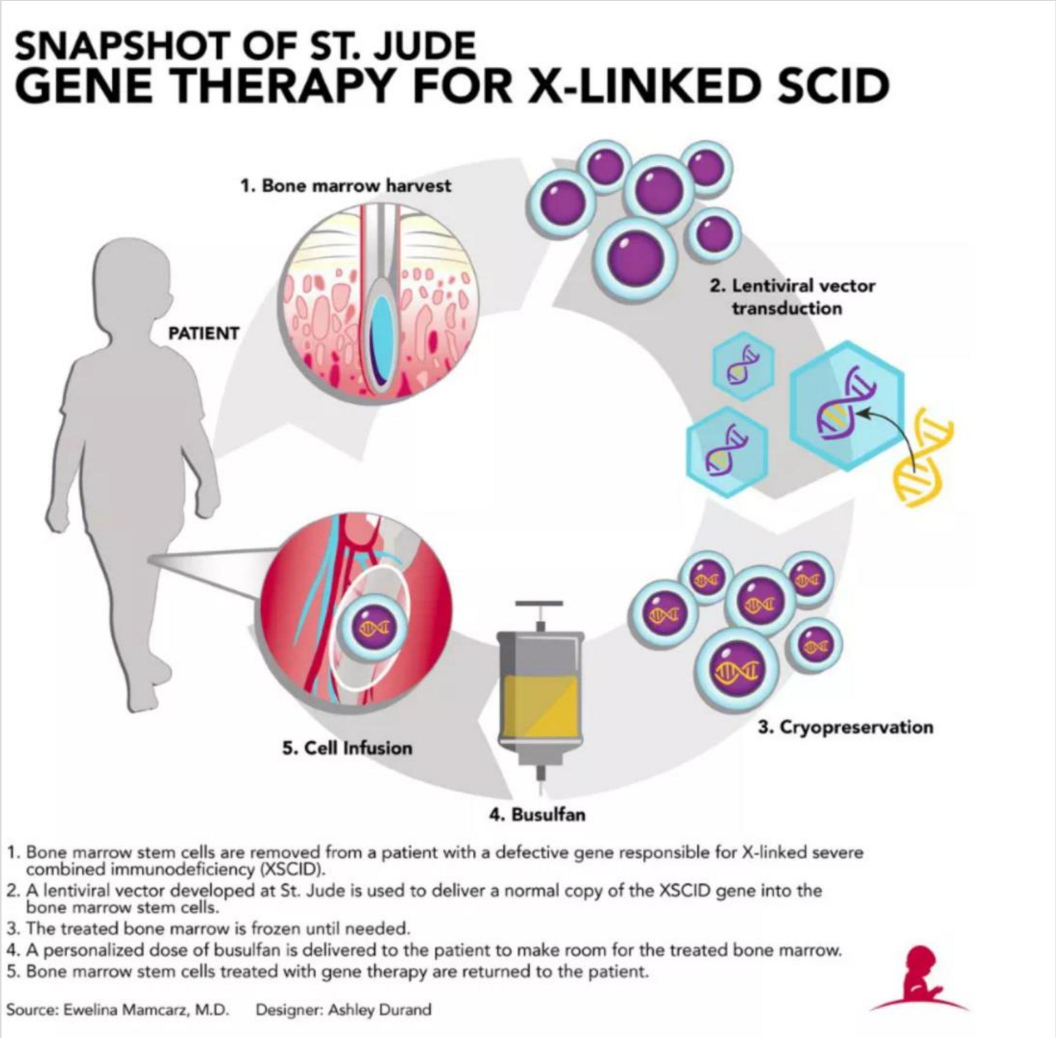
- Bubble baby disease
- Variants of 20 genes involved
  - Defective IL-2R gamma chain (most common) X linked recessive
  - Adenosine Deaminase Deficiency – autosomal recessive
- Presents within first month – 2 months of life
- Loss of cell mediated AND humoral immunity
- Absence of thymic shadow, germinal centers
- T cells absent, very low/undetectable T cell receptor excision circles (TRECs) * on newborn screen
- Diagnosis: flow cytometry – no T cells, B cells can be present but non functional (in X linked), in ADA T, B and NK cells down.
  - In ADA: increased urine deoxyadenosine
- Failure to thrive, chronic diarrhea, thrush
- Recurrent bacterial, fungal, viral, protozoal infections



Combined T and B cell disorder: Severe combined immunodeficiency (SCID)

Treatment:

- Abx, antiviral, antifungal ppx
- Immunoglobin replacement therapies
- Avoid live attenuated vaccines
- Curative:
  - Hematopoietic stem cell transplant
    - Involves conditioning with chemo drugs– makes space for treated stem cells in bone marrow, prevent rejection and incomplete graft
    - May chose not to if fully matched sibling donor or immune system very down or baby <2 months
  - For ADA deficient: enzyme replacement therapy which then takes over function to break down harmful substance
    - Weekly IM injections
    - Temporary
  - Gene therapy
    - Can get away with low dose conditioning
    - Not FDA approved yet – clinical trials going on



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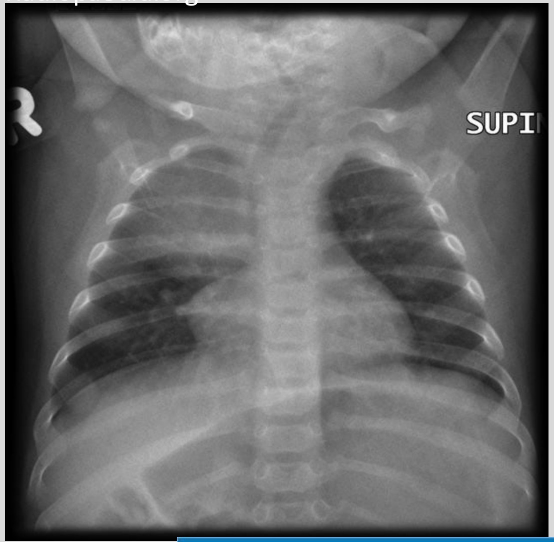
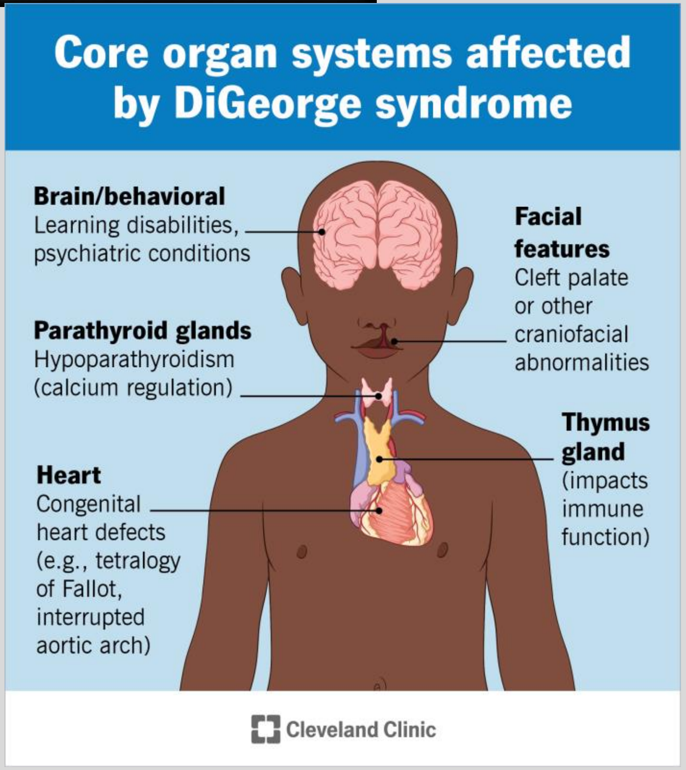
T cell disorder: Thymic aplasia (DiGeorge Syndrome)

- 22q11.2 microdeletion
- Failure of development - 3rd and 4th pharyngeal pouches
    - Absent thymus and parathyroid (immune dysfunction and hypocalcemia)
    - Cleft palate + cleft lip
    - Facial abnormalities – micrognathia
    - Cardiac Defects (tetralogy of Fallot, truncus arteriosus)
- T cells don't get educated, decreased #s in periphery (but B and NK cell #s are fine)
    - Paracortex in lymph nodes will be underdeveloped
- Recurrent viral/fungal infections
- Absent thymic shadow on CXR in neonates
- Treatment:
    - Temporary: ppx antivirals, antibiotics, antifungals
    - One time thymic tissue transplant – FDA approved in 2021

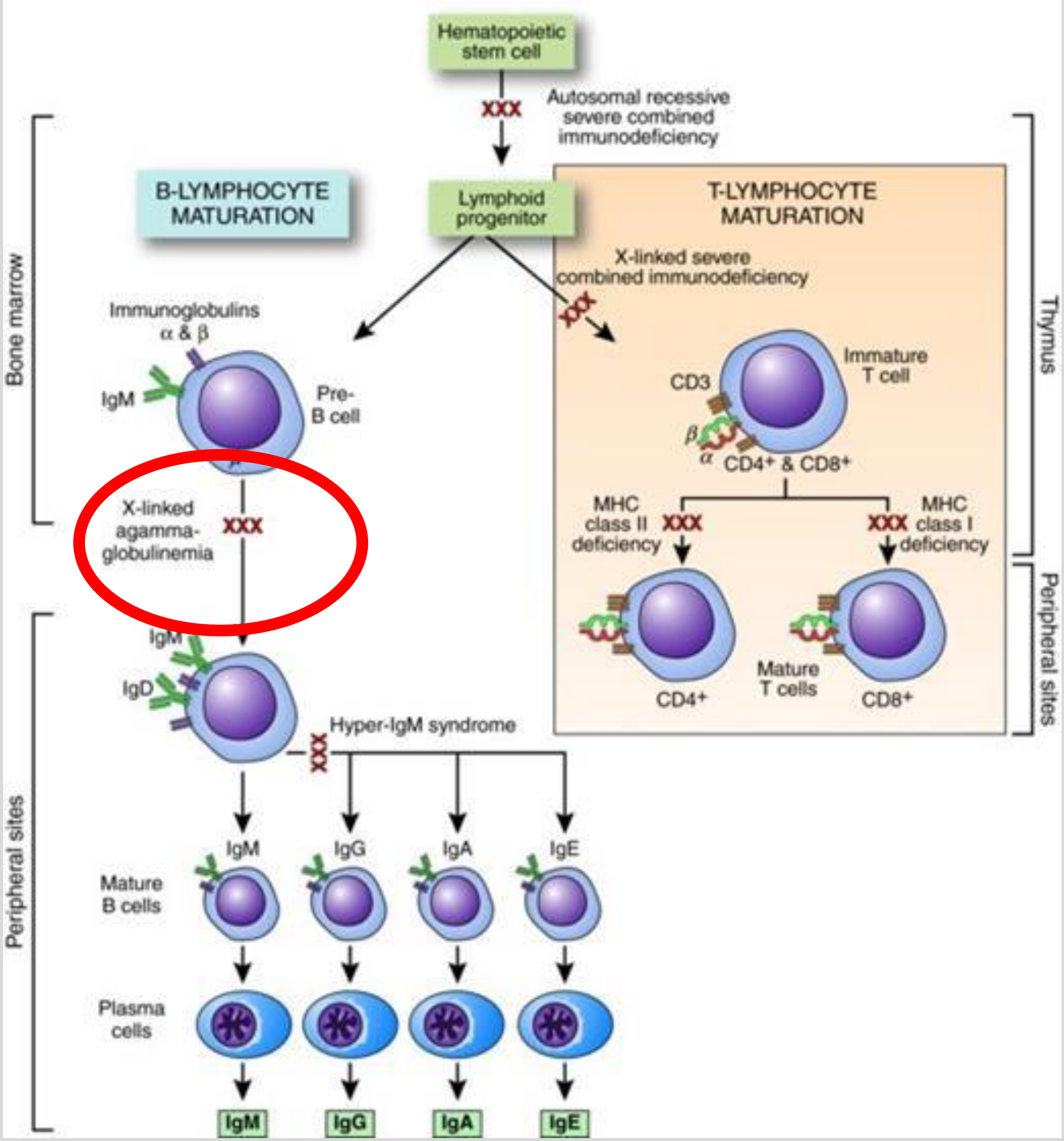
Radiopaedia.org





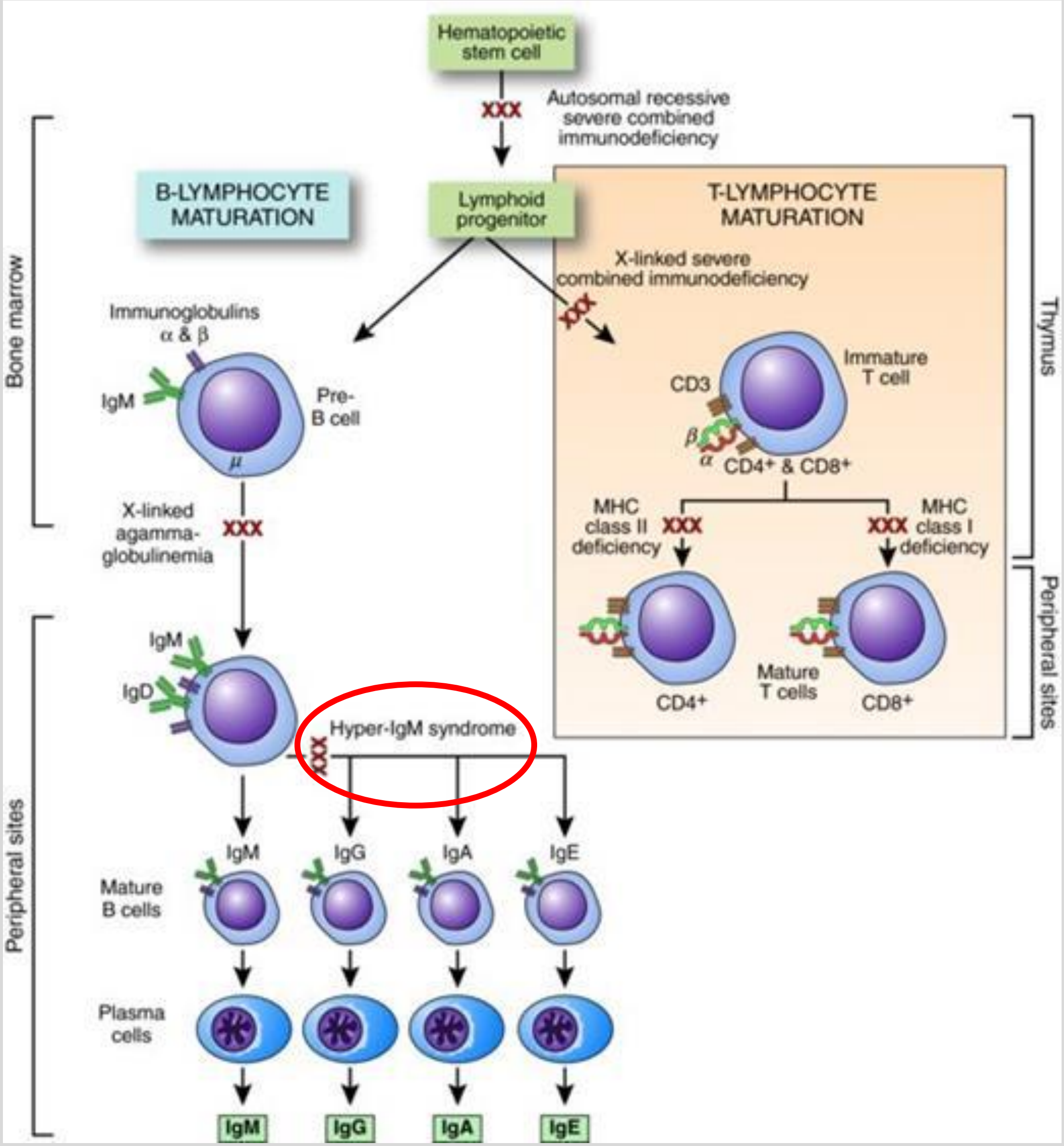
B cell disorder: Bruton X linked agammalobulinemia

- X linked recessive (affecting Boys)
- Defect in BTK gene (tyrosine kinase) which makes light chains in antibodies
- Don't get maturation of B cells
- Absent B cells in peripheral blood!! (CD19+ cells LOW)
    - Trapped in bone marrow
    - Absent lymph nodes/tonsils
    - Primary follicles and germinal centers are absent
- Immunoglobulins all decreased – present after 6 months of age – thanks to mom
- Recurrent bacterial (H flu, Strep Pneumo so recurrent otitis media, sinusitis, pneumonias)
- Enteroviral infections
- Live vaccines contraindicated
- Treatment: lifelong Immunoglobin replacement therapy



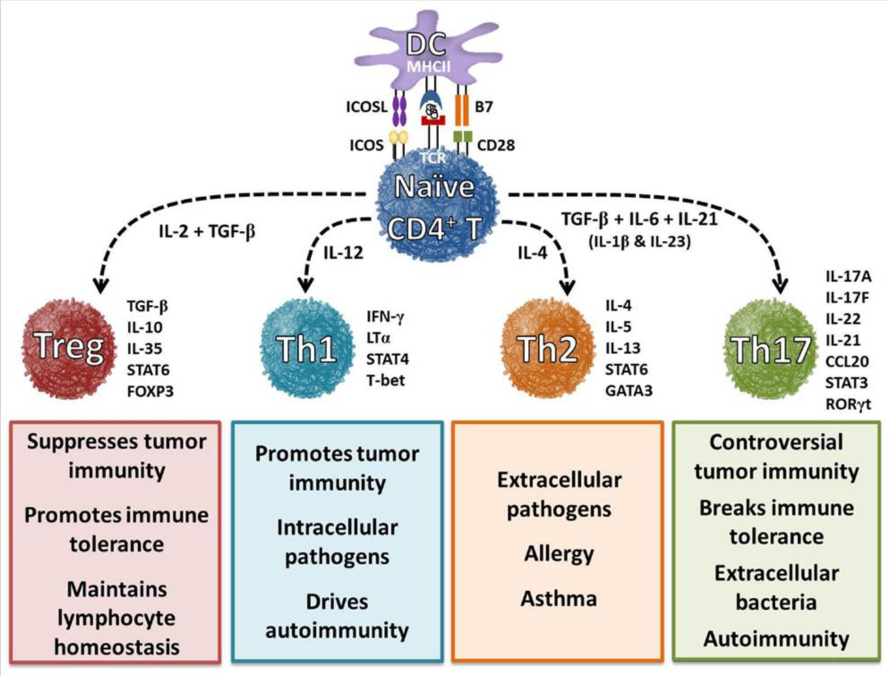
B cell disorder: Hyper IgM syndrome

- X linked recessive
- Can't class switch due to defective CD40L on Th cells
- Normal or elevated IgM, all others very low
- Failure to make germinal centers
- Severe pyogenic infections early in life
    - Opportunistic infections with Pneumocystis, cryptosporidium, CMV



B cell disorder: Job syndrome

- Autosomal dominant
- Also called hyper-IgE syndrome
- Deficiency of Th17 cells due to STAT3 mutation on Ch17
- Impairs recruitment of neutrophils to site of infection
  - Cold Staphylococcal Abscesses
  - Retained Baby Teeth
  - Coarse facies
  - Dermatologic problems (eczema)
  - IgE increased, eosinophils increased
  - Fractures from minor trauma

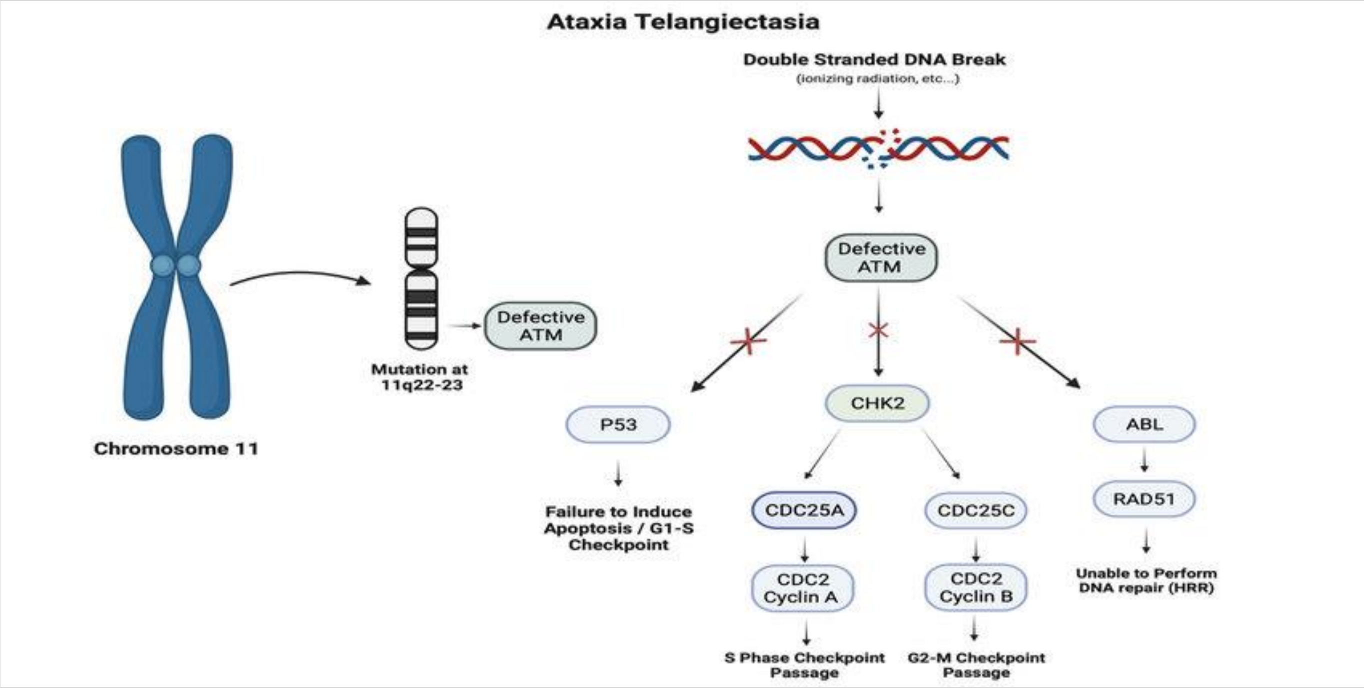


Other combined: Wiskott-Aldrich syndrome

- X linked recessive
- Mutation in *WAS* gene
  - Affects leukocytes (B and T cell function) + platelets
  - Unable to reorganize actin cytoskeleton
  - Important in antigen presentation
- Presents with recurrent pyogenic infections (sinus, ear, pneumonia)
- Thrombocytopenia (fewer, smaller PLTs) - bleeding common cause of death
- Eczema
- Increased risk of autoimmune disease and malignancy
- Decreased to normal IgG, IgM levels, increased IgE and IgA levels
- Avoiding live attenuated vaccines
- Tx: Abx ppx, IV immunoglobulins
- Previously bone marrow transplant after conditioning
  - FDA approved 2025 – gene therapy via viral vector –start making WASp

Other combined: Ataxia-telangiectasia (A-T)

- Autosomal Recessive (AR)
- Defects in *ATM* gene (on chromosome 11q22.3
  - Failure to detect DNA manage
  - Failure to stop progression of cell cycle
  - Prone to more mutations
- Onset of symptoms 2-5 years of age, decline may not be noticeable until 4-8 years of age
- Cerebellar Defects (causing ataxia, dysarthria, dysphagia)
- Spider Angiomas (telangiectasia)
  - Cutaneous
  - Ocular
- IgA deficiency
- Increased sensitivity to radiation (limit xray exposure)
- At risk of lymphoma and leukemia
- Increased AFP, IgG and IgE, T cell depression variable
- Lymphopenia, cerebellar atrophy -> leads to uncoordinated movements + falls
- Treatment:
  - Immunoglobulin therapy
  - Ppx Abx



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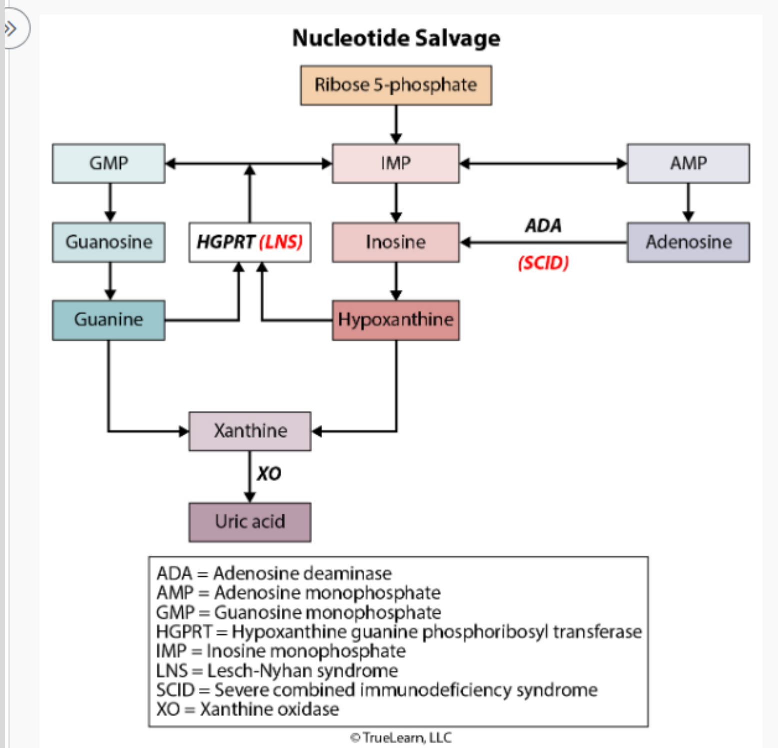
Starter Question

- A 2 month old M infant is brought into your office with a recent hospital admission for 2 weeks for Pneumocystis (PCP) Pneumonia requiring respiratory support, in reviewing the chart, the infant has had multiple episodes of thrush that required prolonged antifungal treatment. Growth chart shows the infants head circumference, weight and length are below the <1.5%tile, on exam, he has severe seborrheic dermatitis and a bulging R TM (tympanic membrane). Parent notes baby has 10 epsiodes of watery diarrhea every day. You review the xray from the hospital and they comment on an absent thymic shadow. Flow Cytometry shows decreased B, T and NK cells and decreased immunoglobulins. What is the most likely mechanism causing this presentation? What about the Newborn screening show?

- Disease process described: SCID (severe combined immunodeficiency)
  - Two flavors – X linked IL-2R mutation or AR - ADA deficient
  - This is a male patient however all three cells (B,T,NK) are decreased – which goes along more with ADA deficiency
  - Chronic diarrhea, Failure to thrive, recurrent viral, fungal, bacterial, protozoal infections
  - NBS would show very low T cells – or TERCs



Severe Combined Immunodeficiency	
Pathogenesis	• X-linked: defective interleukin-2 receptor γ chain • Autosomal recessive: adenosine deaminase deficiency
Presentation	• < 3 months of age • Recurrent bacterial, viral, fungal, and protozoal infections • Opportunistic infections (eg, *Pneumocystis jirovecii*) • Failure to thrive • Chronic diarrhea • Thrush
Diagnosis	• No thymic shadow in chest x-ray • ↓↓ lymphocytic count • Low levels of all immunoglobulins • Flow cytometry: low T and B cells
Management	• Treatment of infections with antibiotics • Immunoglobulin replacement therapy • Curative treatment: bone marrow transplant

Another question for retention

- 5 year old M brought in to the ED for 3 days of severe L ear pain associated with fever. You check his medical history and note he has had 4 episodes of AOM this year despite having ear tubes placed by ENT when he was 3 years of age. When you complete a med reconciliation, you note he just completed a 5 day course of sulfamethoxazole/trimethoprim for a skin infection that was superimposed on his eczematous lesions, as per mother, it has not improved. You suspect there's more than just another AOM going on and get some basic blood work. What do you want to start with?
    - CBC with differential, IgG levels, plus/minus CXR, Bcx depending on how the kid looks, Immunology consult
    - Say the labs show PLT 50k (nml 150-400k) IgM 20 (nml 40-230) IgA 500 (70-400 is nml), serum IgG 900 (nml 700-1600)
    - What is the most likely defect associated with this immunodeficiency?

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A) microdeletion of long arm of chromosome 22

B) mutation of the adenosine deaminase gene on chromosome 20

C) mutation of the Bruton tyrosine kinase gene on the X chromosome

D) mutation of the signal transducer and activator of transcription 3 gene on chromosome 17

E) mutation of the wiskott-aldrich syndrome gene on the X chromosome

A) microdeletion of long arm of chromosome 22 - DIGEORGE

B) mutation of the adenosine deaminase gene on chromosome 20 - SCID

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D) mutation of the signal transducer and activator of transcription 3 gene on chromosome 17 – STAT3 mutation- Job syndrome

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E) mutation of the wiskott-aldrich syndrome gene on the X chromosome – Wiskott-Aldrich syndrome (WAS)

Explanation

- WAS protein used to link signaling proteins to actin cytoskeleton reorganization in hematopoietic cells.
- Defective T cell function because impairment of migration and adhesion. Number of mature B cells reduced, function dysregulated. NK cell function also defective
- Thrombocytopenia - due to decreased production, increased destruction and intrinsic PLT abnormalities
- Labs: PLT 20-50k, smaller than normal
- Decreased number and function of T cells
- Decreased Absolute lymphocyte count
- Abnormal IG subtypes: low IgM, high IgA and IgE, normal or low IgG
- VS Job syndrome (STAT3 gene mutation on Ch17) severe eczema, pulm infections, coarse facies, retained baby teeth, bone fractures, eosinophilia, significantly elevated IgE, also recurrent skin staph infections without pus formation

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Sources:

Nelson's Textbook of Pediatrics

Zitelli and Davis' Atlas of Pediatric Physical Diagnosis

Immune Deficiency Foundation

Step up to Pediatrics

Rubin's Pathology

Current Diagnosis and Treatment – pediatrics

Watch – MedLecturesMadeEasy – Primary Immunodeficiency disorders
Or Osmosis lectures on immunodeficiencies

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Thank you for listening!

Lecture Feedback:

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